

EDO de Primer orden

$$y' + y \tan x = \cos^2 x, \quad y(0) = 1$$

$$P(x) = \tan(x)$$

$$u = e^{\int \tan(x) dx} = e^{-\ln \cos x} = \cancel{e^{\ln \cos x}^{-1}} = \cos(x)^{-1}$$

$$\frac{y'}{\cos x} + y \frac{\tan x}{\cos x} = \frac{\cos^2 x}{\cos x}$$

$$\frac{d}{dx} \left[\frac{1}{\cos x} \cdot y \right] = \cos x$$

$$\int \frac{d}{dx} \left[\frac{1}{\cos x} \cdot y \right] = \int \cos x \, dx$$

$$\frac{y}{\cos x} = \sin x + C_1$$

Solución:

$$\frac{y}{\cos x} = \sin x + 1$$

$$y = \sin(x) \cos(x) + \cos(x)$$

Aplicamos C.I: $y(0) = 1$

$$\frac{1}{\cancel{\cos(0)}} = \cancel{\sin(0)} + C_1$$

$$1 = C_1$$

EDO de Segundo orden:

$$y'' + 4y = x^2$$

$$y(D^2 + 4) = x^2$$

Homogénea:

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm 2i$$

No Homogénea:

$$x^2$$

$$\boxed{D^3}$$

aniquilador

$$m = 0_3$$

$$y_c = C_1 \cos(2x) + C_2 \sin(2x)$$

$$y = y_c + A e^{0x} + Bx e^{0x} + Cx^2 e^{0x}$$

$$y = y_c + A + Bx + Cx^2$$

$$y = A + Bx + Cx^2$$

$$y' = B + 2Cx$$

$$y'' = 2C$$

Substitución:

$$2C + 4(A + Bx + Cx^2) = x^2$$

$$2C + 4A + 4Bx + 4Cx^2 = x^2$$

x^2 :

$$4C = 1$$

$$C = \frac{1}{4}$$

x :

$$4B = 0$$

$$B = 0$$

Constante:

$$2C + 4A = 0$$

$$2\left(\frac{1}{4}\right) + 4A = 0$$

$$A = -\frac{1}{8}$$

$$y_p = \frac{1}{4}x^2 - \frac{1}{8}$$

Solución general:

$$y = C_1 \cos(2x) + C_2 \sin(2x) + \frac{1}{4}x^2 - \frac{1}{8}$$

EDO de 2×2 :

$$u'(t) = Au(t) + f(t), \quad A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \quad f(t) = \begin{pmatrix} t \\ e^{-t} \end{pmatrix}$$

$$u_1'(t) = 2u_1(t) + u_2(t) + t$$

$$u_2'(t) = 2u_2(t) + e^{-t}$$

$$u_2' - 2u_2 = e^{-t}$$

$$p(t) = -2 \quad u(t) = e^{\int -2dt} = e^{-2t}$$

$$e^{-2t} \cdot u_2 = \int e^{-3t}$$

$$e^{-2t} u_2 = -\frac{1}{3} e^{-3t} + C_1$$

$$u_2 = -\frac{1}{3} e^{-t} + C_1 e^{2t}$$

Sustitución de u_2 en ec.1:

$$u_1' = 2u_1 + \left(-\frac{1}{3} e^{-t} + C_1 e^{2t}\right) + t$$

$$u_1' - 2u_1 = -\frac{1}{3} e^{-t} + C_1 e^{2t} + t$$

$$p(t) = -2$$

$$u(t) = e^{\int -2 dt} = e^{-2t}$$

$$e^{-2t} u_1 = \int \left(-\frac{1}{3} e^{-3t} + C_1 + t e^{-2t} \right) dt$$

$$\begin{array}{c} + \quad \uparrow \quad \text{D} \\ - \quad \downarrow \quad \text{I} \\ + \quad \downarrow \quad \text{I} \end{array} \begin{array}{c} t \\ 1 \\ 0 \end{array} \begin{array}{c} e^{-2t} \\ -e^{-1t}/2 \\ e^{-2t}/4 \end{array} \uparrow = -\frac{1}{2} t e^{-2t} - \frac{1}{4} e^{-2t}$$

$$e^{-2t} u_1 = \frac{1}{9} e^{-3t} + C_1 t - \frac{1}{2} t e^{-2t} - \frac{1}{4} e^{-2t} + C_2$$

$$u_1 = \frac{1}{9} e^{-t} + C_1 t e^{2t} - \frac{1}{2} t - \frac{1}{4} + C_2 e^{2t}$$

Solución general:

$$u(t) = \begin{pmatrix} \frac{1}{9} e^{-t} + C_1 t e^{2t} - \frac{1}{2} t - \frac{1}{4} + C_2 e^{2t} \\ -\frac{1}{3} e^{-t} + C_1 e^{2t} \end{pmatrix}$$